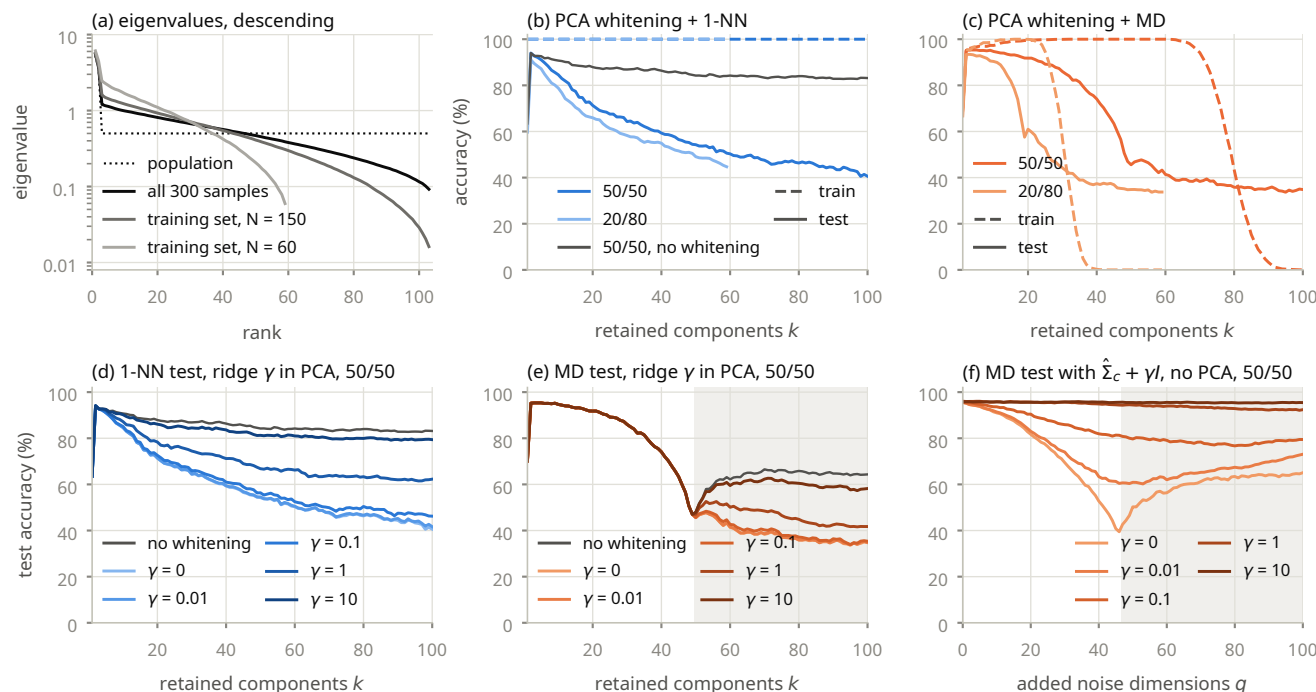
(a) eigenvalues, descending
eigenvalue
rank
population
all 300 samples
training set, N = 150
training set, N = 60
(b) PCA whitening + 1-NN
accuracy (%)
retained components k
50/50
20/80
50/50, no whitening
train
test
(c) PCA whitening + MD
retained components k
50/50
20/80
train
test
(d) 1-NN test, ridge γ in PCA, 50/50
test accuracy (%)
retained components k
no whitening
γ = 0
γ = 0.01
γ = 0.1
γ = 1
γ = 10
(e) MD test, ridge γ in PCA, 50/50
retained components k
no whitening
γ = 0
γ = 0.01
γ = 0.1
γ = 1
γ = 10
(f) MD test with $\hat{\Sigma}_c + \gamma I$, no PCA, 50/50
added noise dimensions q
γ = 0
γ = 0.01
γ = 0.1
γ = 1
γ = 10